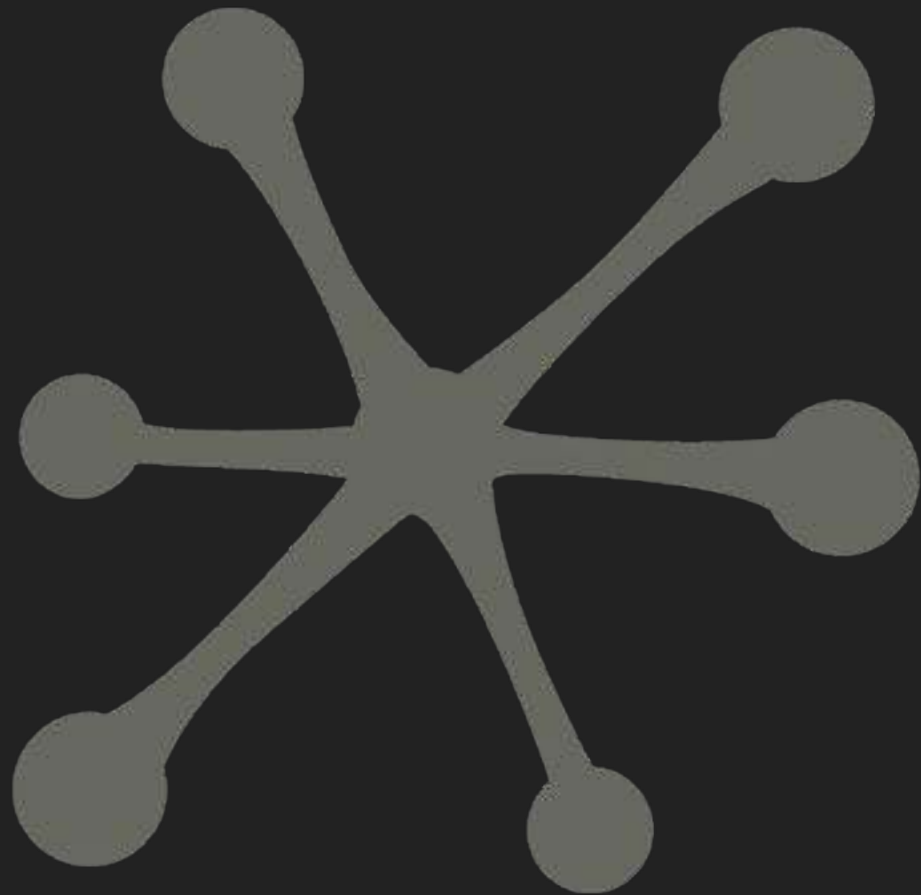
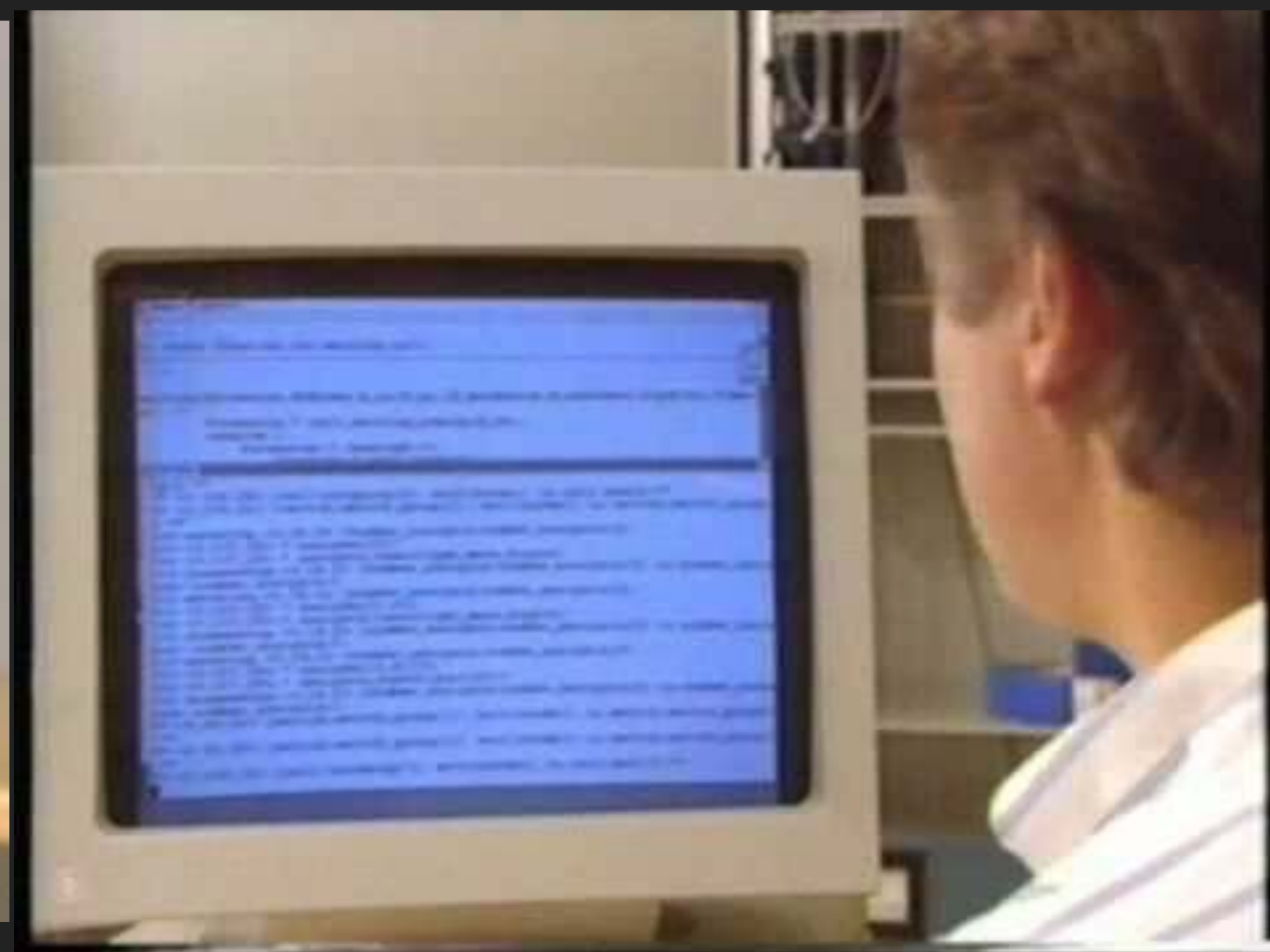




ENHANCED EMBEDDED SYSTEMS

NERVES PROJECT

1980



Easy to build projects for everyone

Everyday ELECTRONICS

OCT. 8
70p

FREE

AND INSIDE A
CHOICE OF DESIGN
TO BUILD ON THIS
STRIPBOARD

CAREER or PASTIME



**SMALLER!
EASIER!**



Monochrome (1.101w 07-May-08) (Last on Wed May 14 13:36)



MONOCHROME
New streamlined layout! **Easier to use!** New files! **Extra exclamation marks!**

Dish some dirt at <MT0> today!

~~~~~ archon ~~~

Menu [ESC] = Utilities (inc. Talker & **EXIT**)

**You don't use ssh. Booo!**

Menu [I] = Help and **I**nformation on Monochrome

**Welcome to  
the new  
version of  
Monochrome!  
(version 1.101w)**

Menu [N] = **N**ews and Media

Menu [T] = Science, Technology and Medicine

Menu [E] = **E**ntertainment

Menu [C] = Society and **C**ulture

Menu [R] = **R**ecreation

Menu [M] = **M**onochrome Users

Hello 'SexDrugs&DrumMachinesForAgRaveGeneration'. (evilandi:4)

<< 22 other users at Sun Jan 11 19:30 BST >> █





## Version 1.5 - Install Disk for PC's

Insert this disk in your floppy drive (A or B); set the DOS prompt to the correct drive (A or B), and type **INSTALL** to boot.

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(00000001)

JUST USE  
NEW VERSION  
TO TRY AN





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10 results ▾

Google Search

I'm feeling lucky

*Index contains ~25 million pages (soon to be much bigger)*

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[Stanford Search](#) [Linux Search](#)

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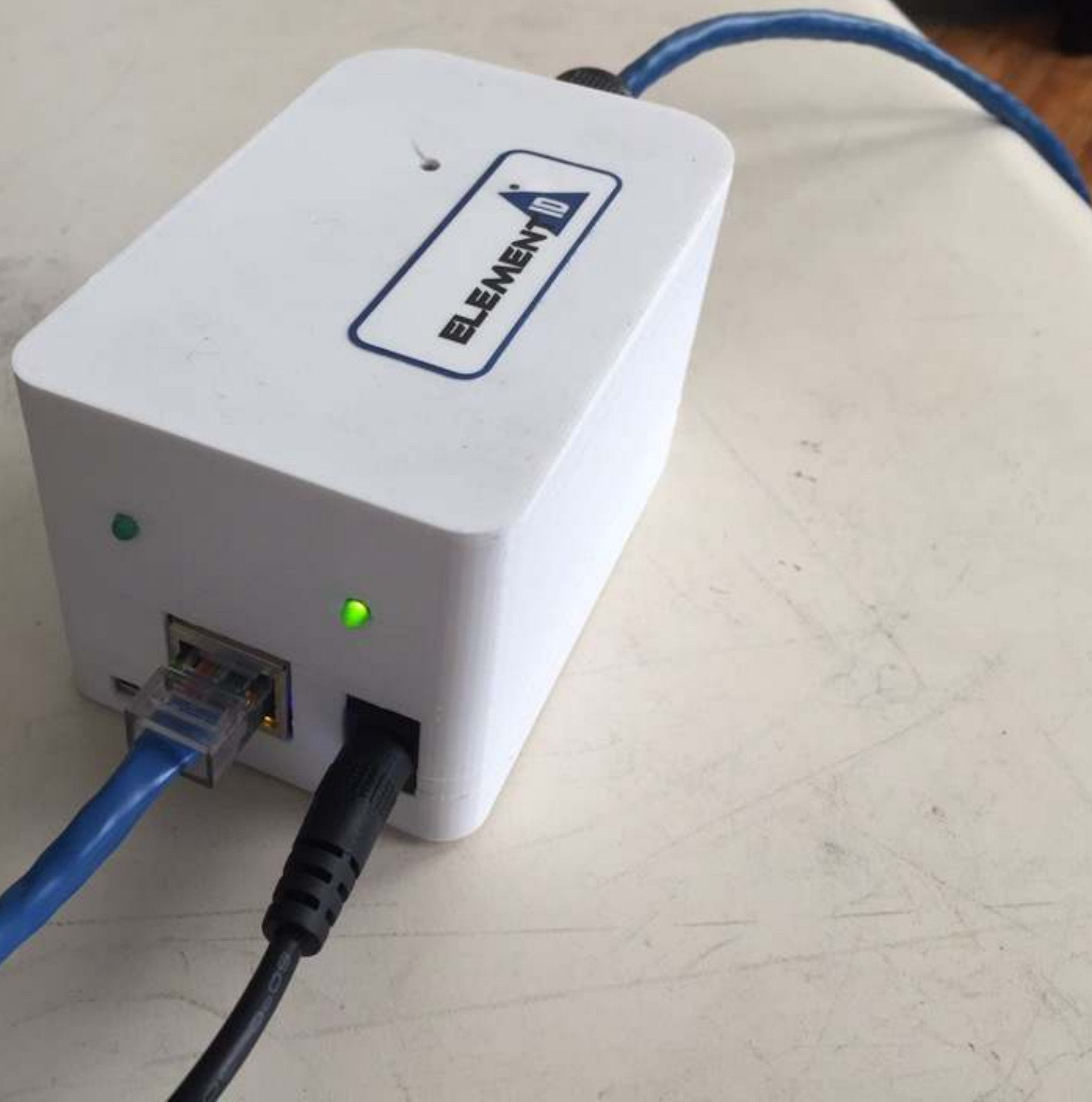
**Phoenix**  
Framework

**I NEED AN EMBEDDED SYSTEM  
AND I NEED IT IN 1 WEEK.**

Element ID

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**NERVES LET US CREATE AND  
DELIVER A PRODUCTION PRODUCT  
IN 5 DAYS WITHOUT SACRIFICING  
PERFORMANCE OR RELIABILITY.**

**Element ID**

**...UNLIKE OTHER EMBEDDED PLATFORMS  
WE'VE USED, MODIFICATIONS AND  
FEATURE ENHANCEMENTS ARE GOING TO  
BE EASY TO DO IN THE FUTURE.**

**Element ID**



2 Web Developers

5 Days



Frank  
Hunleth

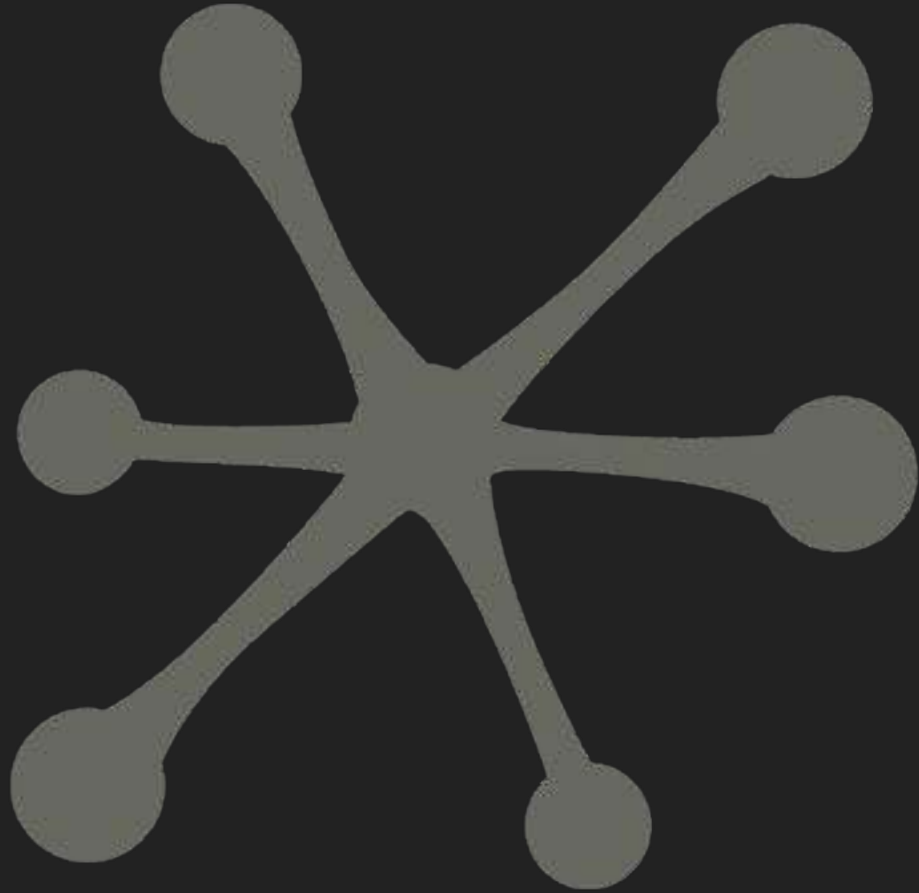


Justin  
Schneck



Garth  
Hitchens

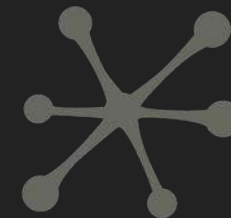
Community  
elixir-lang slack #nerves



WHAT IS

---

**NERVES**

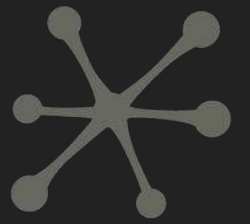


**FRAMEWORKS**

**PLATFORM**

**TOOLING**





## FRAMEWORKS

nerves\_led

nerves\_networking

nerves\_uart

elixir\_ale

nerves\_io\_neopixel

nerves\_ssdp\_server

nerves\_ssdp\_client

nerves\_hub

## PLATFORM

nerves\_system\_ag150

nerves\_system\_alix

nerves\_system\_bbb

nerves\_system\_rpi

nerves\_system\_rpi2

nerves\_system\_rpi3

nerves\_system\_br

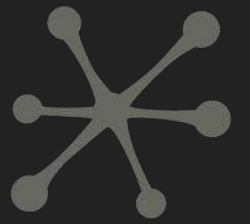
nerves\_toolchain

nerves\_toolchain\_...

## TOOLING

mix tasks

- mix nerves.new
- mix nerves.loadpaths
- mix nerves.precompile
- mix firmware
- mix firmware.burn utilities
- fwup
- cell



## SUPPORTED TARGETS

## TARGET NAME

Raspberry Pi B / A+ /B+ / Zero

rpi

Raspberry Pi 2

rpi2

Raspberry Pi 3

rpi3

BeagleBone Black

bbb

Alix

alix

AG150

ag150

Intel Galileo 2

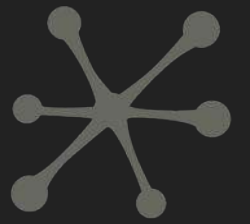
galileo

Lego EV3

ev3

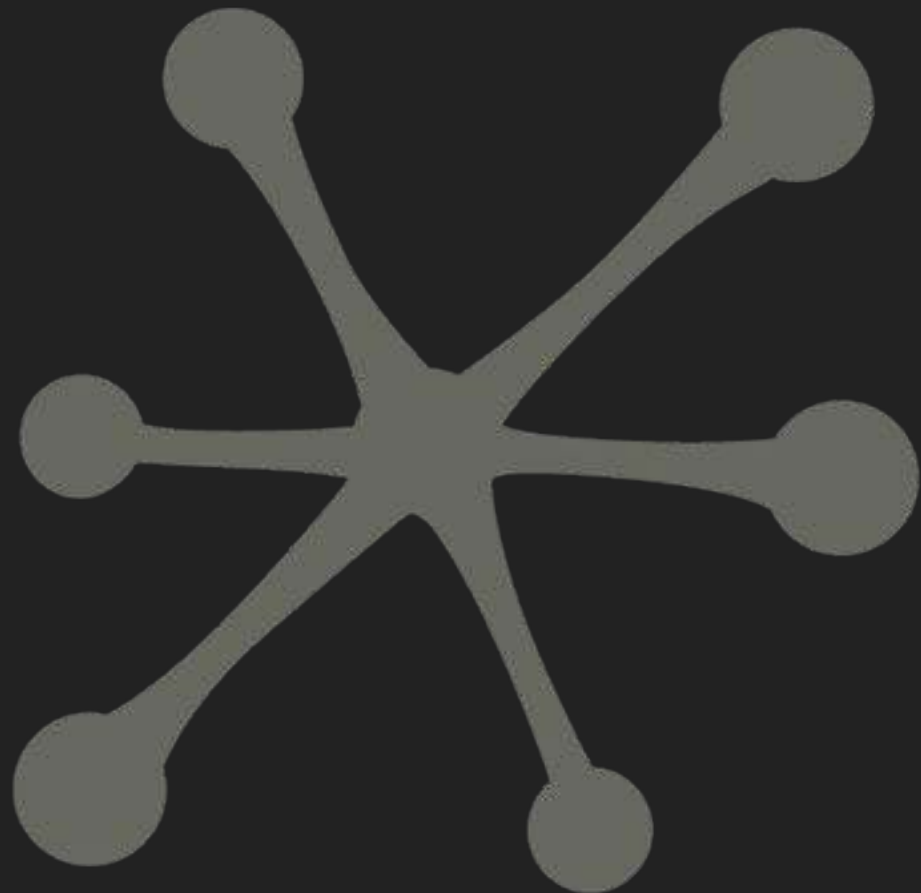
QEmu Arm

qemu\_arm



## LINUX SINGLE BOARD COMPUTERS

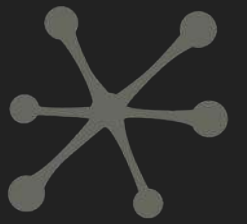




GETTING STARTED

---

**SANDBOX**

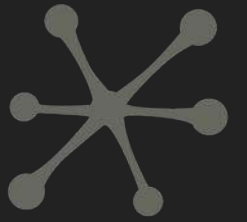


## RASBIAN / DEBIAN LINUX

- Update system
- Establish network
- SSH
- Install Erlang (ESL)
- Install Elixir
- Checkout Blinky
- mix run



THE SANDBOX

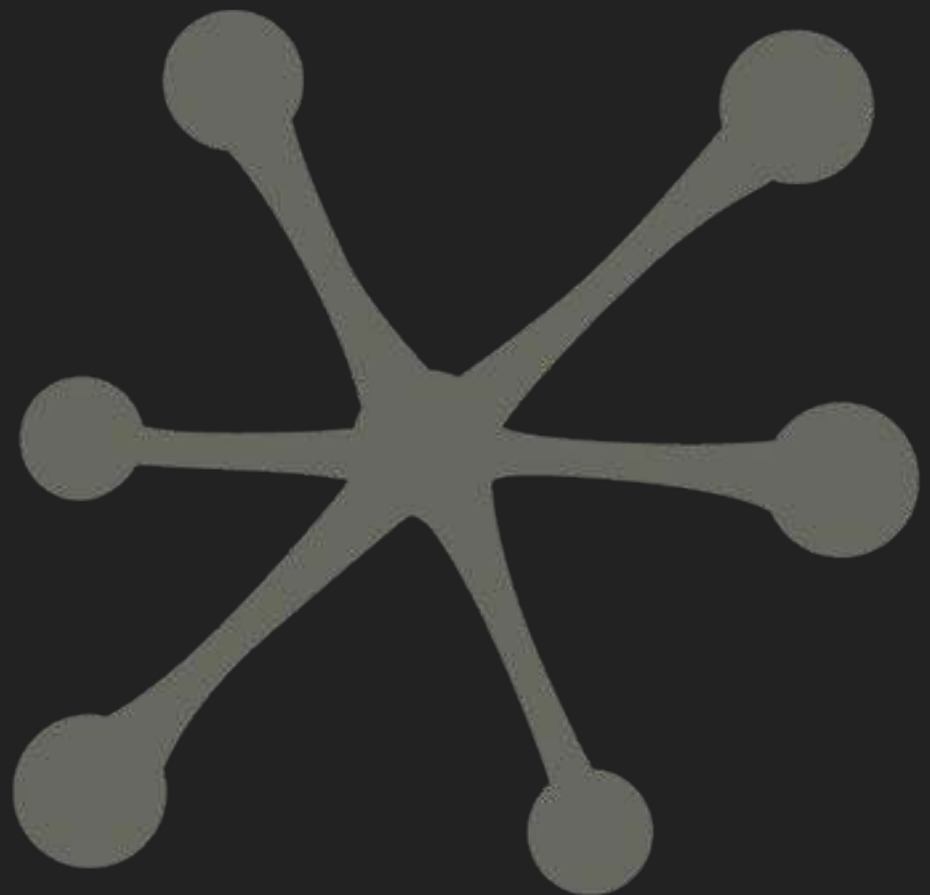


---

# RASBIAN / DEBIAN LINUX



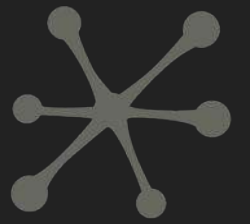
**THERE HAS TO  
BE A BETTER  
WAY!**



GETTING STARTED

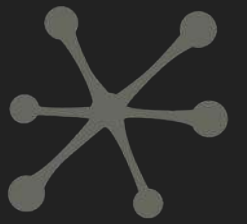
---

**NERVES PLATFORM**



# LETS MAKE THIS EASY

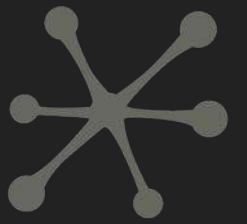
```
# install bake
Bakefile...
bake system get -target
bake toolchain get -target
bake firmware
bake burn
```



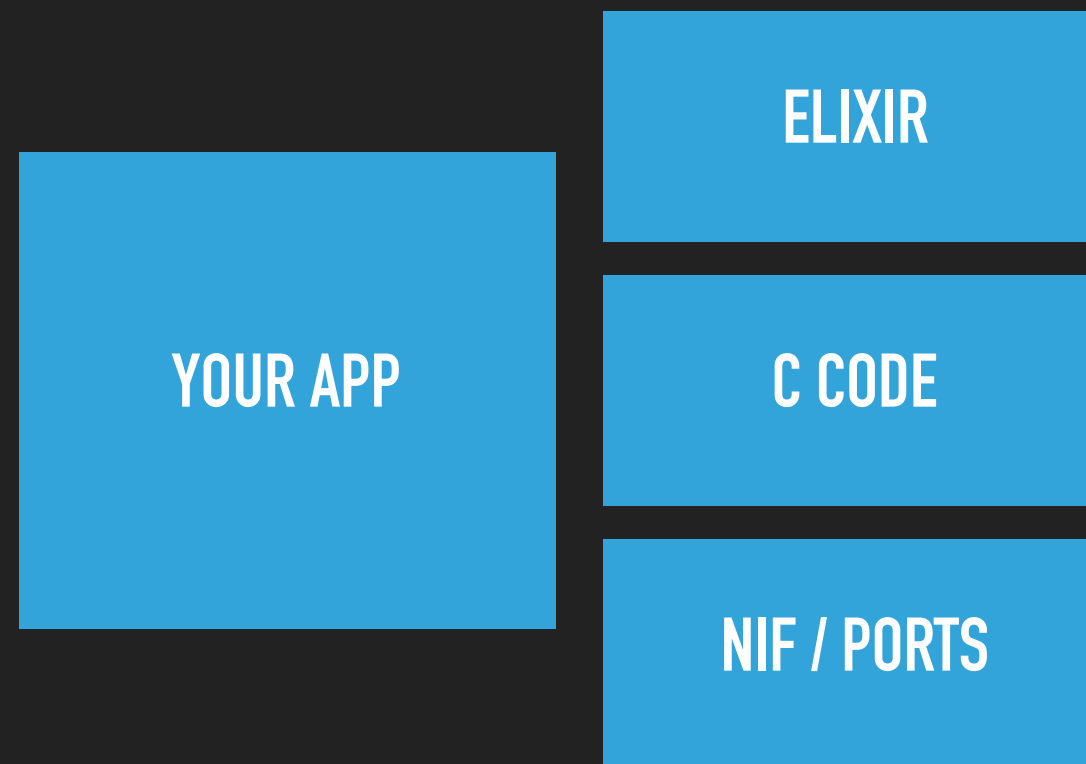
## LETS MAKE THIS EASY

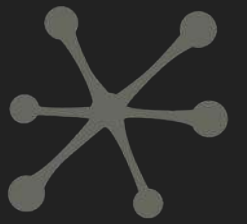
```
mix deps.get  
mix firmware  
mix firmware.burn
```



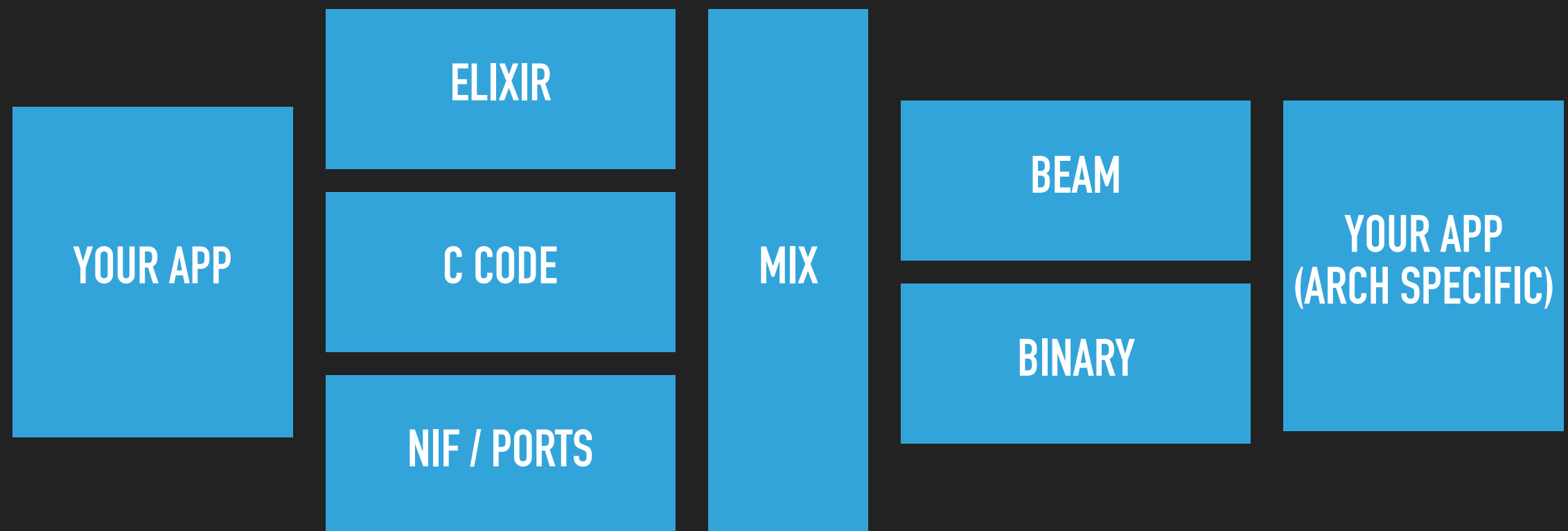


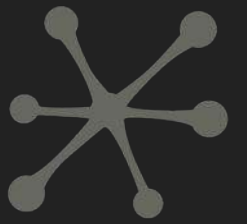
# MIXING FIRMWARE



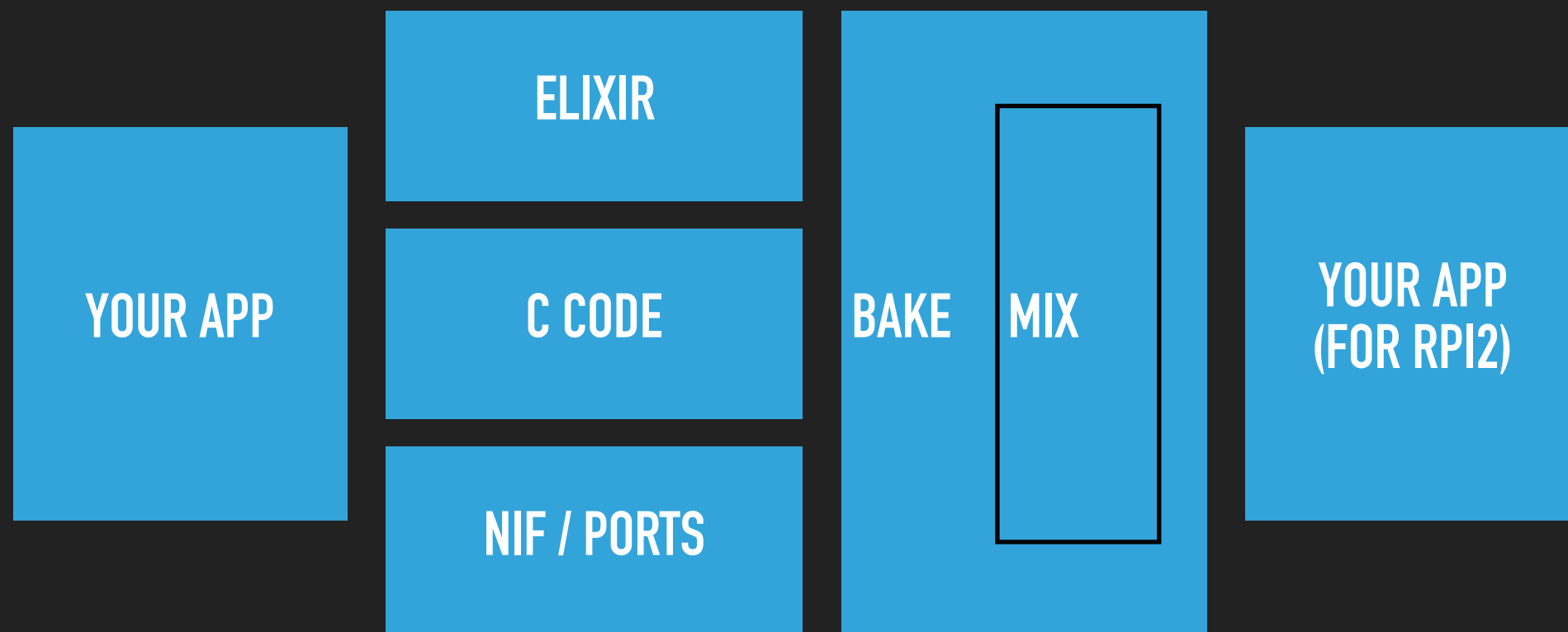


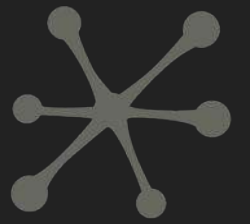
## COMPILING ON YOUR MACHINE



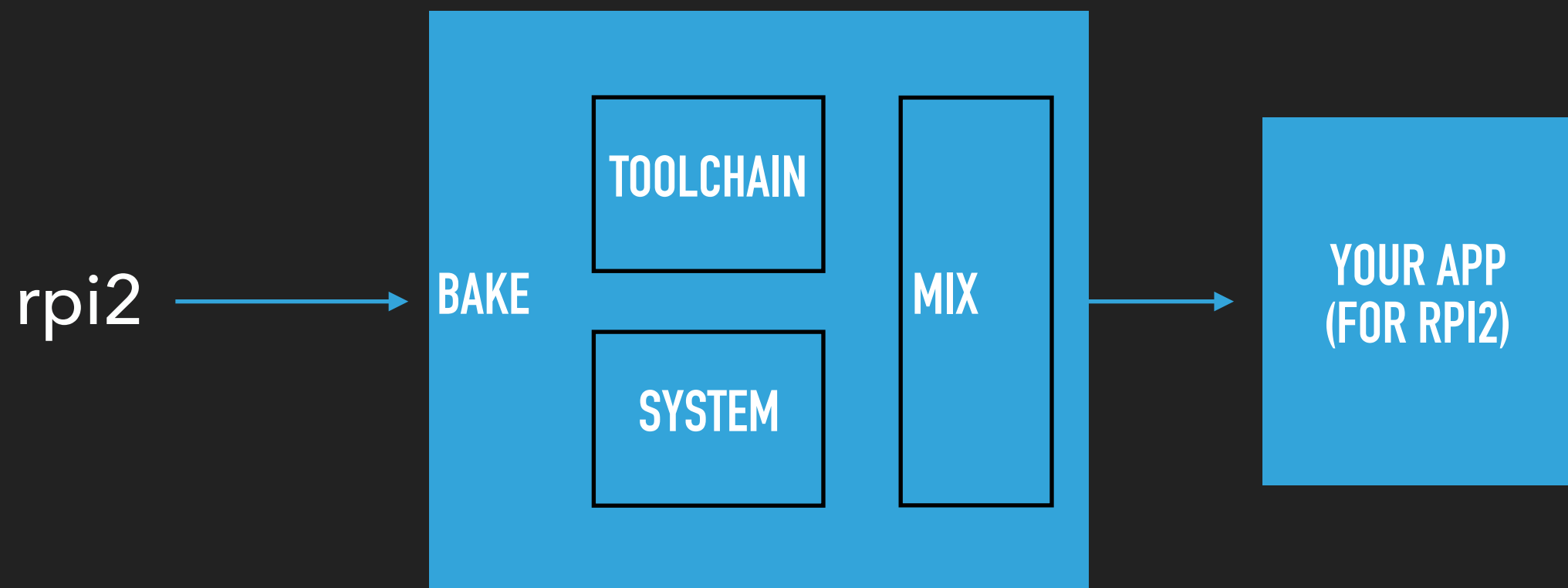


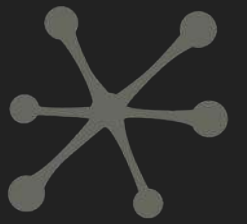
## MIXING FIRMWARE WITH BAKE



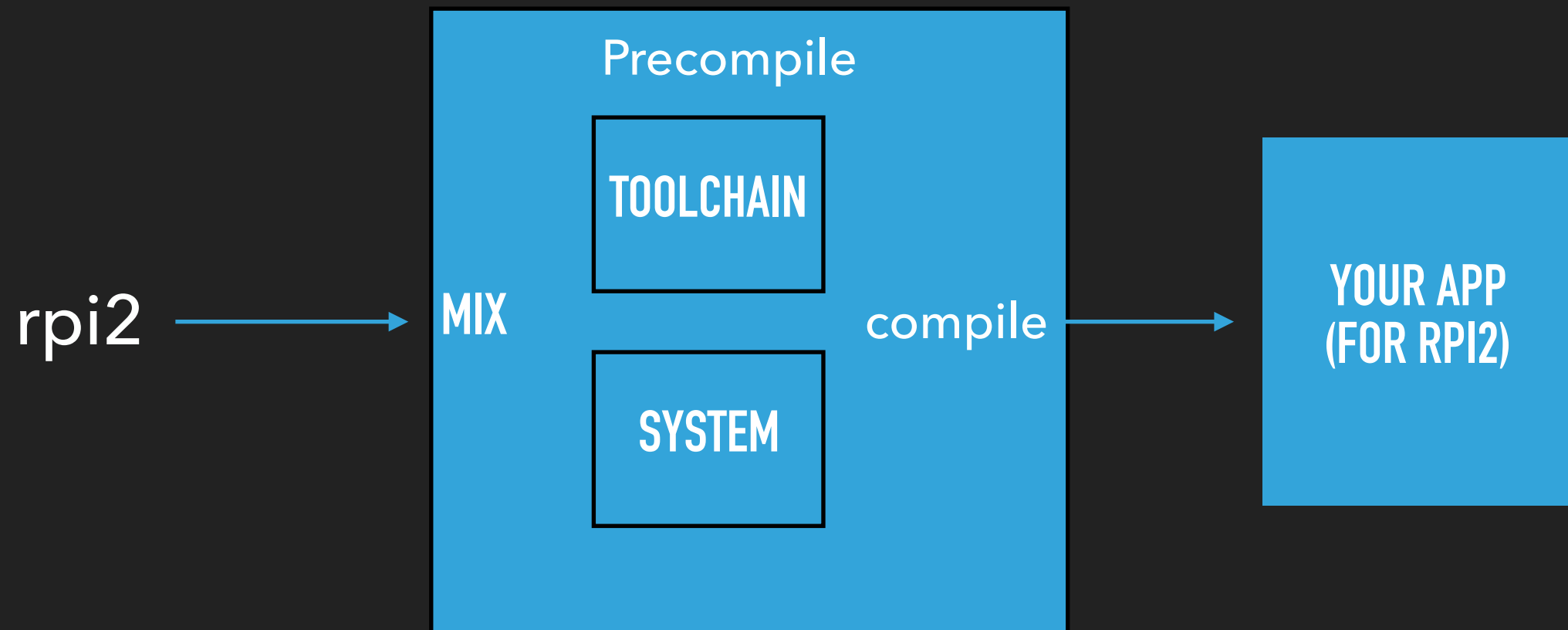


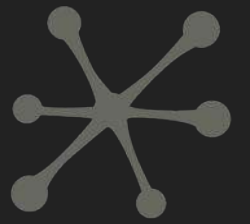
## MIXING FIRMWARE WITH BAKE





## MIXING FIRMWARE





## TOOLCHAINS

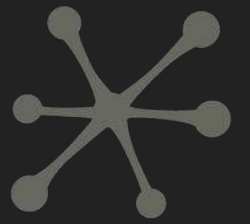
### TOOLCHAIN CONFIG

- crosstool
- for target
- host configs

### TOOLCHAIN

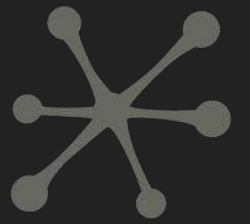
- compilers
- run on host
- compile for target





## TOOLCHAIN CONFIG

```
CT_LOCAL_TARBALLS_DIR="${CT_TOP_DIR}/../dl"
CT_SAVE_TARBALLS=y
CT_PREFIX_DIR="${CT_TOP_DIR}/../x-tools/${CT_TARGET}"
# CT_REMOVE_DOCS is not set
CT_LOG_EXTRA=y
CT_ARCH_FLOAT_HW=y
CT_ARCH_arm=y
CT_KERNEL_linux=y
CT_KERNEL_V_3_4=y
CT_BINUTILS_LINKER_LD_GOLD=y
CT_BINUTILS_GOLD_THREADS=y
CT_BINUTILS_LD_WRAPPER=y
CT_BINUTILS_PLUGINS=y
...
```



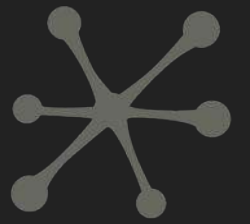
## SYSTEMS

### SYSTEM CONFIG

- buildroot
- defconfig
- rootfs-  
additions

### SYSTEM

- bootfoles
- rootfs
- linux  
kernel



## SYSTEM CONFIG

BR2\_arm=y

BR2\_cortex\_a7=y

BR2\_ARM\_FPU\_NEON\_VFPV4=y

BR2\_TOOLCHAIN\_EXTERNAL=y

BR2\_TOOLCHAIN\_EXTERNAL\_CUSTOM=y

BR2\_TOOLCHAIN\_EXTERNAL\_DOWNLOAD=y

BR2\_PACKAGE\_NERVES\_CONFIG\_APPS="crypto"

BR2\_PACKAGE\_NERVES\_CONFIG\_EXTRA\_MOUNTS="/dev/mmcblk0p3:/root:vfat

BR2\_PACKAGE\_NERVES\_CONFIG\_HANG\_ON\_EXIT=y

BR2\_PACKAGE\_NERVES\_CONFIG\_UNIQUEID\_PROG="\\"/usr/bin/boardid -b rp4\\""

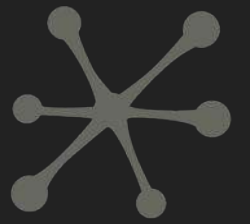
BR2\_PACKAGE\_NERVES\_CONFIG\_HOSTNAME\_PATTERN="nerves-%.4s"

BR2\_PACKAGE\_HOST\_ERLANG\_RELSYNC=y

BR2\_PACKAGE\_HOST\_ERLANG\_RELX=y

BR2\_PACKAGE\_HOST\_FWUP=y

...



## MIX LIFECYCLE

MIX      deps.precompile

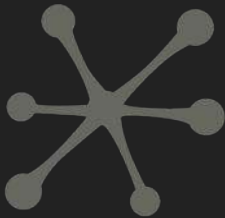
BOOTSTRAP

nerves\_system

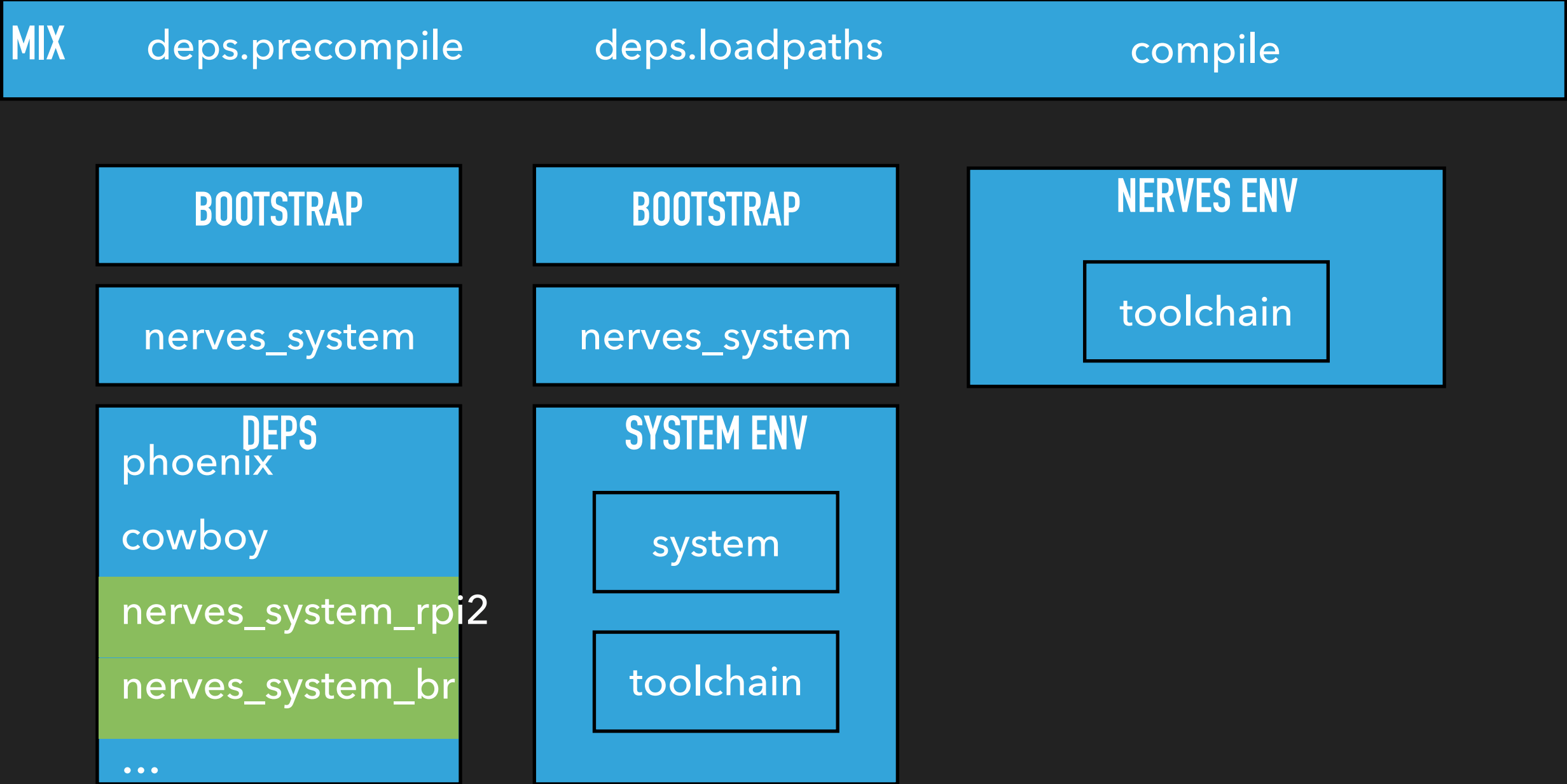
DEPS  
phoenix  
cowboy  
nerves\_system\_rpi2  
nerves\_system\_br  
...

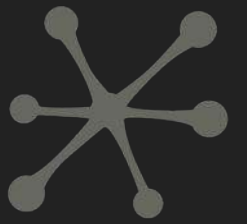
nerves.exs

```
config :nerves_system_rpi2, :nerves_env,  
  type: :system,  
  mirrors: [  
    "https://github.com/nerves-project/  
nerves_system_rpi2/releases/download/  
v#{version}/nerves_system_rpi2-  
v#{version}.tar.gz"],  
  build_platform: Nerves.System.Platforms.BR,  
  build_config: [  
    defconfig: "nerves_defconfig"  
  ]
```



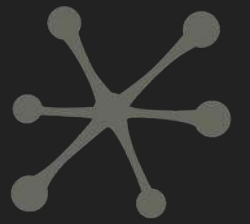
# MIX LIFECYCLE





# NERVES BOOTSTRAP

```
mix archive.install https://github.com/nerves-project/archives/raw/master/nerves\_bootstrap.ez
```

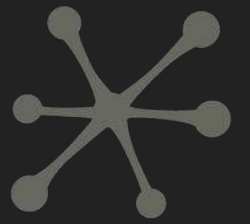


### MIX FILE

```
defmodule Blinky.Mixfile do
  use Mix.Project

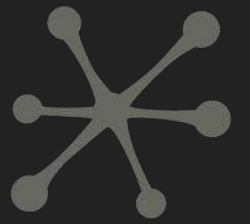
  @target System.get_env("NERVES_TARGET") || "rpi2"
  ...
end
```





## MIX FILE

```
defmodule Blinky.Mixfile do
  ...
  def project do
    [app: :blinky,
     version: "0.1.0",
     archives: [nerves_bootstrap: "~> 0.1"],
     target: @target,
     deps_path: "deps/#{@target}",
     build_path: "_build/#{@target}",
     config_path: "config/#{@target}/config.exs",
     aliases: aliases,
     deps: deps ++ system(@target)]
  end
end
```



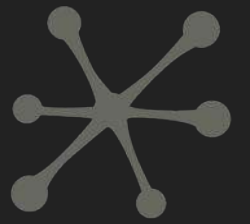
## MIX FILE

```
defmodule Blinky.Mixfile do
  ...

  def system("rpi2") do
    [{:nerves_system_rpi2, "~> 0.4.0"}]
  end

  def aliases do
    ["deps.precompile": ["nerves.precompile", "deps.precompile"],
     "deps.loadpaths": ["deps.loadpaths", "nerves.loadpaths"]]
  end

end
```



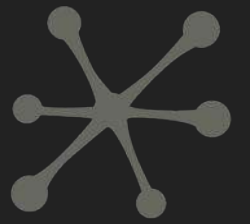
## MIX FILE

```
defmodule Blinky.Mixfile do
  ...
  def system("rpi") do
    [[:nerves_system_rpi, "~> 0.4.0"]]
  end

  def system("rpi2") do
    [[:nerves_system_rpi2, "~> 0.4.0"]]
  end

  def system("rpi3") do
    [[:nerves_system_rpi3, "~> 0.4.0"]]
  end

  ...
end
```

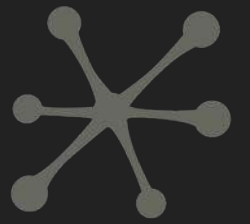


## CHANGING TARGETS

```
NERVES_TARGET=rpi3 mix deps.get
```

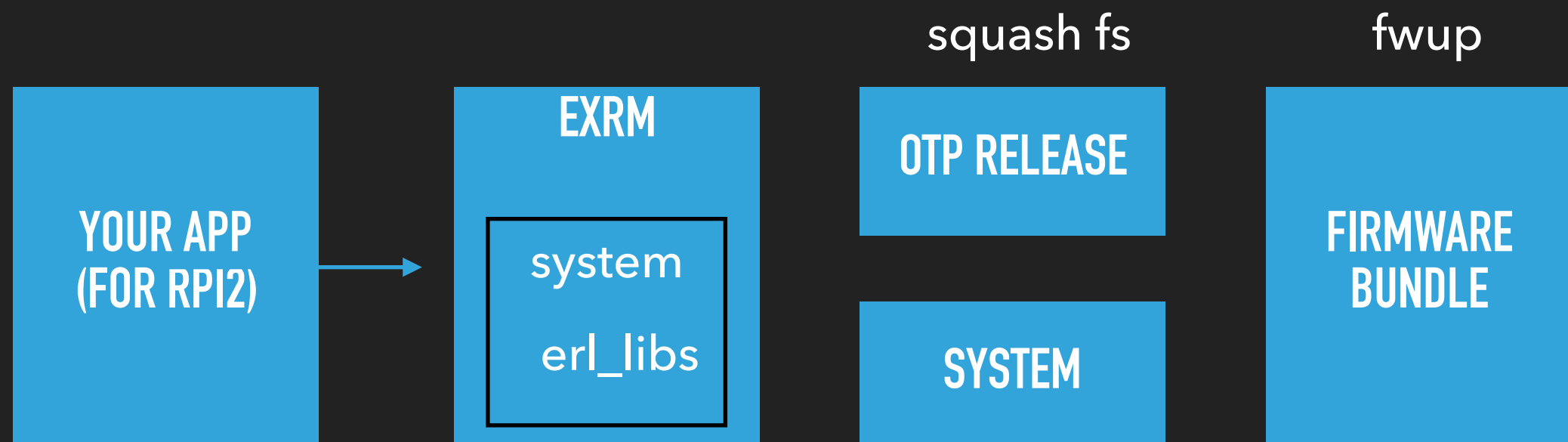
```
export NERVES_TARGET=rpi3  
mix deps.get
```

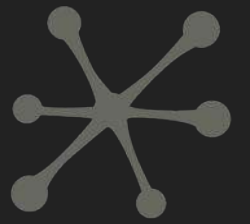
```
mix deps.get  
# @target System.get_env("NERVES_TARGET") || "rpi2"
```



## MIX FIRMWARE

MIX ... compile firmware





## MIX FIRMWARE

MIX ... compile firmware firmware.burn

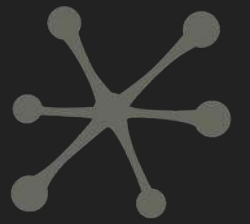
fwup

fwup

YOUR APP  
(FOR RPI2)

FIRMWARE  
BUNDLE

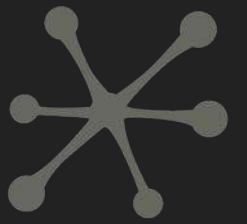
SD CARD/  
FIRMWARE  
IMAGE



## THE RESULT



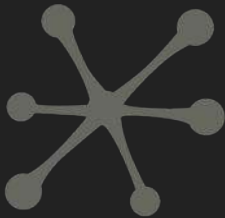




## THE RESULT

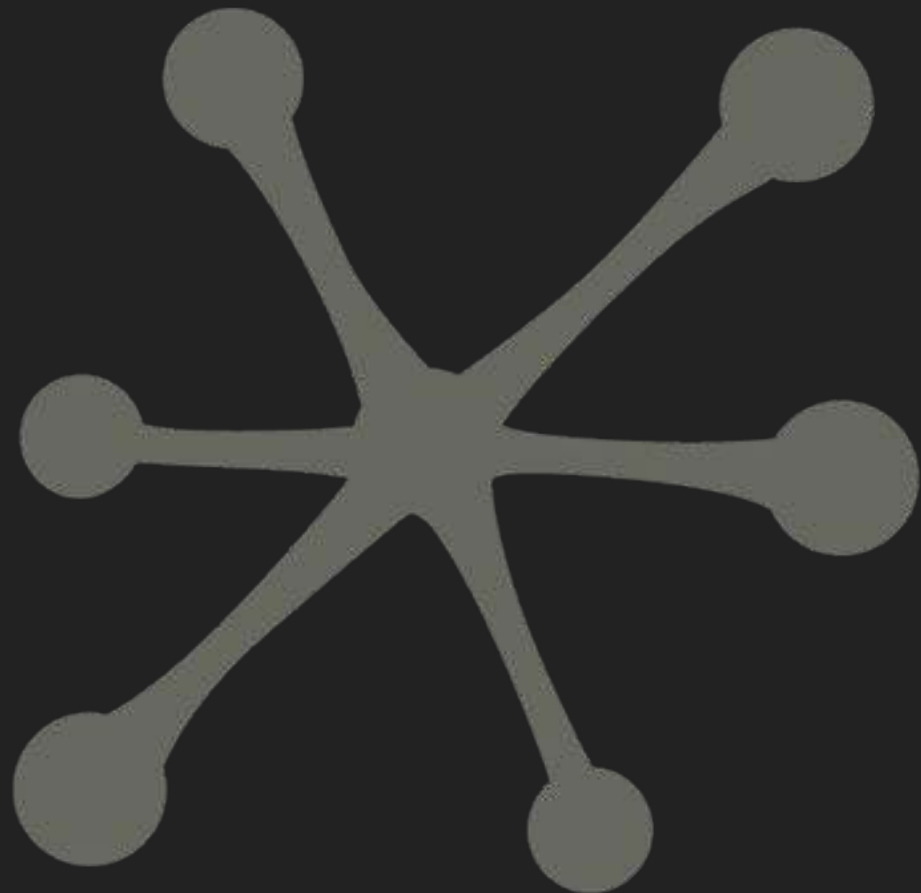






# THE RESULT

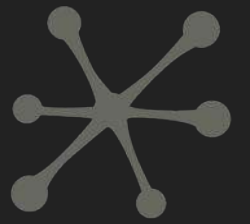




GETTING STARTED

---

# NERVES FRAMEWORK

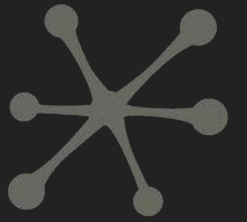


## NETWORKING

```
{:nerves_networking, "~> 0.5.0"}
```

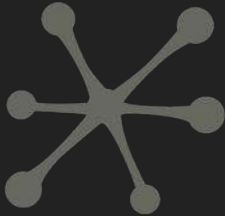
```
{:ok, _} = Networking.setup :eth0
```

```
mode: "static",  
ip: "10.0.0.5",  
router: "10.0.0.1",  
mask: "16",  
subnet: "255.255.0.0",  
mode: "static",  
dns: "8.8.8.8 8.8.4.4",  
hostname: "myhost"
```

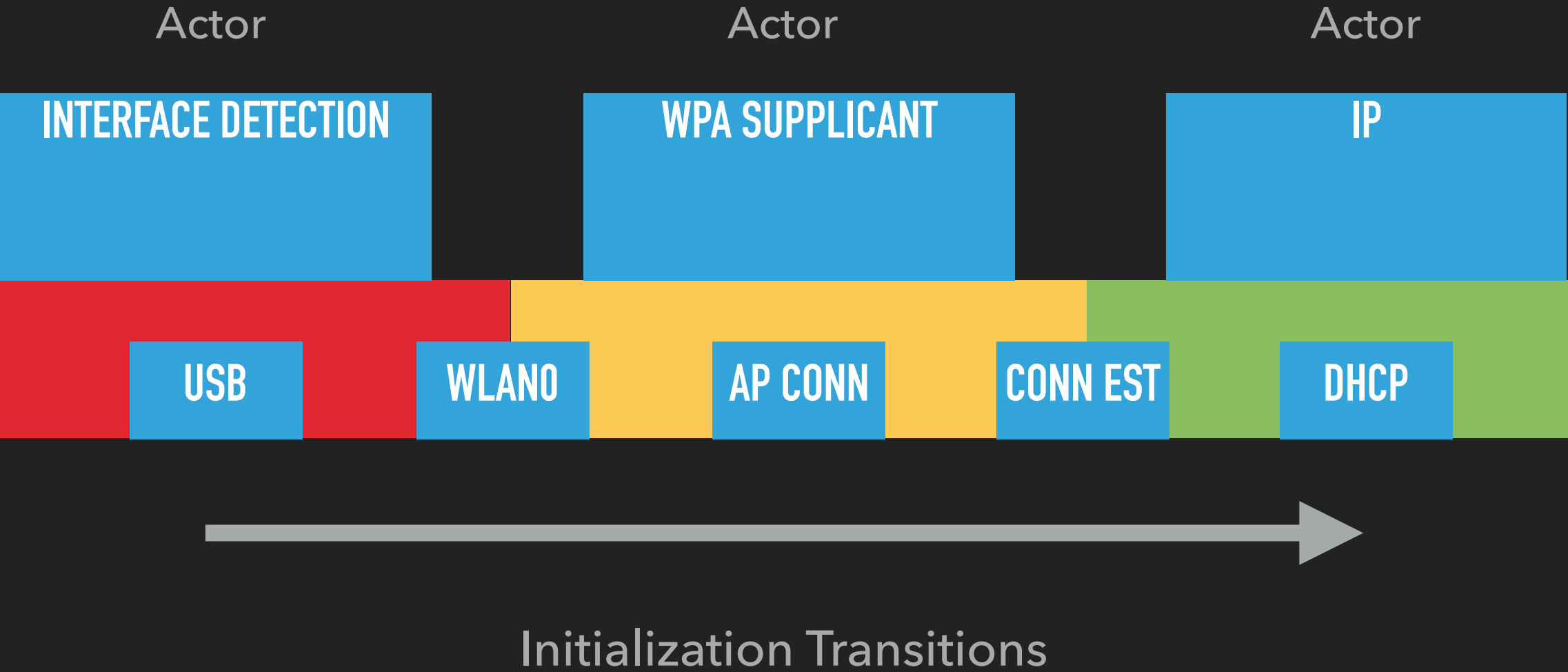


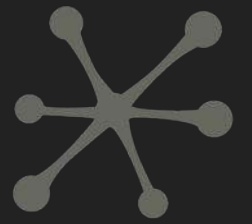
## WIFI

```
{:nerves_wifi, "~> 0.1.0"}
```



WIFI



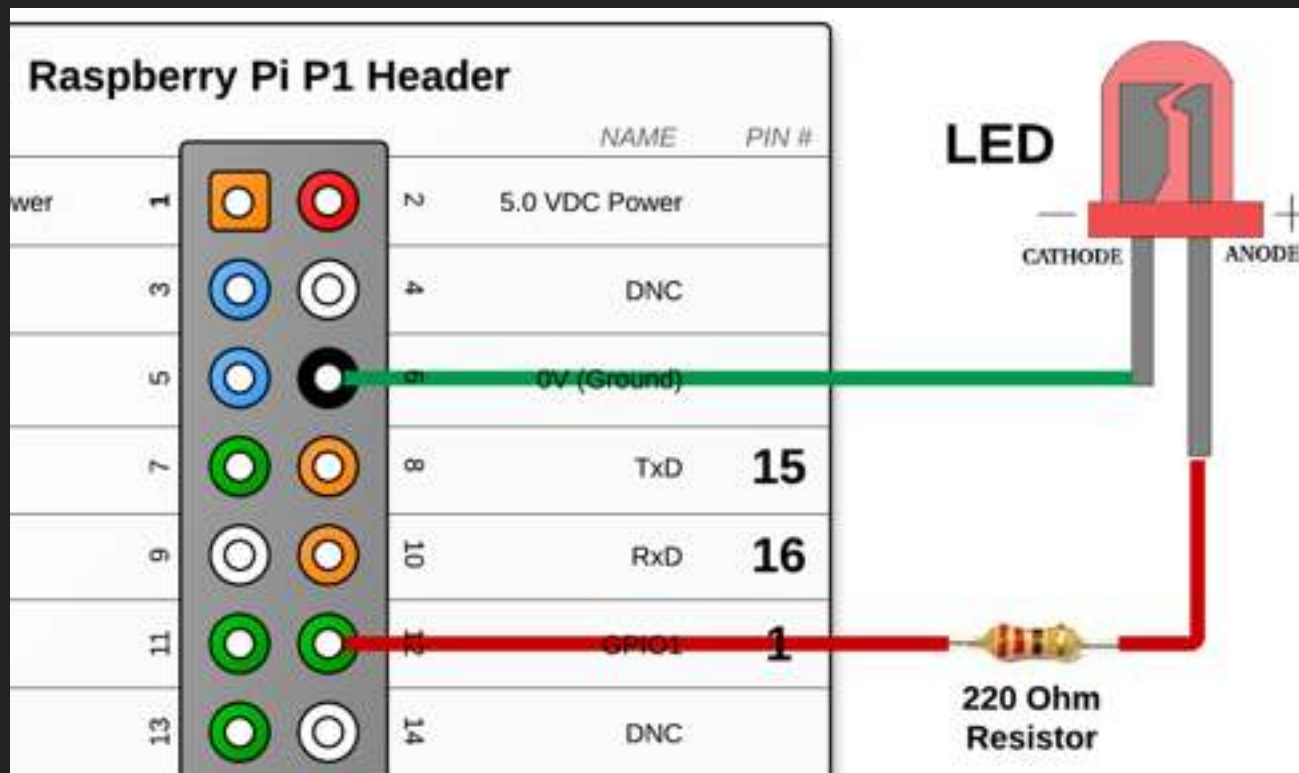


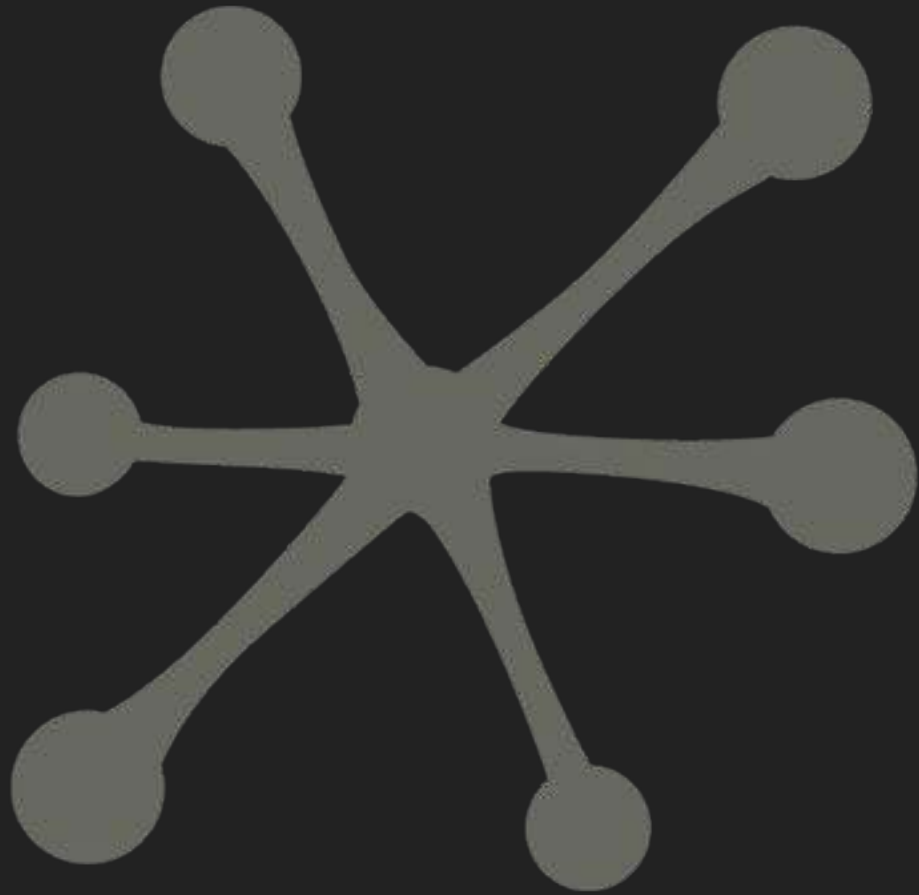
## INTERACTING WITH HARDWARE

```
{:elixir_ale, "~> 0.4.0"}
```

```
{:ok, pid} = Gpio.start_link(1, :output)
```

```
Gpio.write(pid, 1)
```



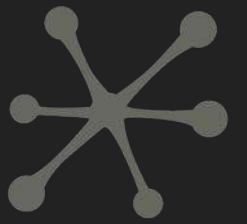


GETTING STARTED

---

# USER INTERFACES





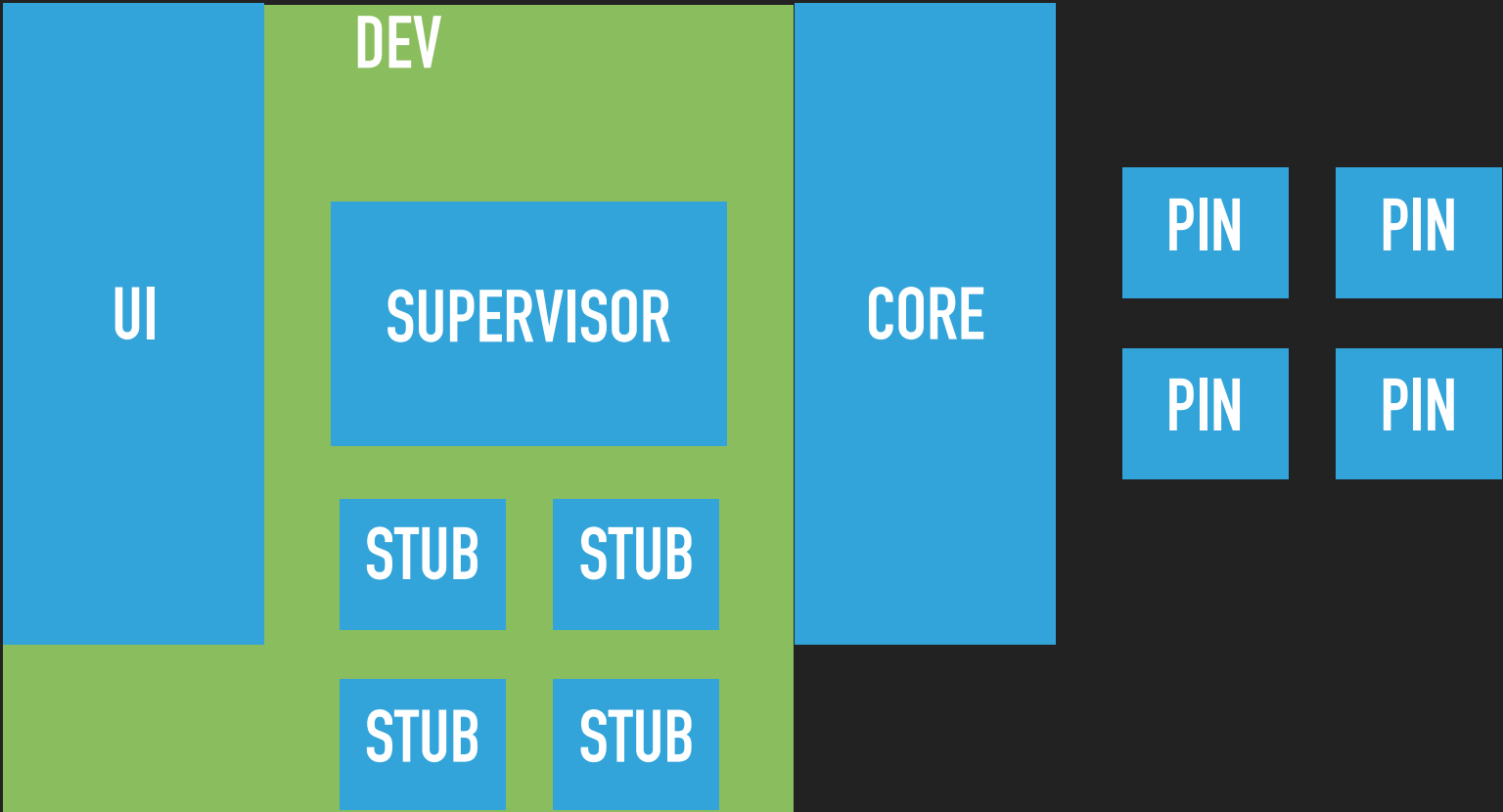
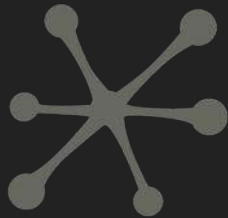
## PHOENIX FOR WEB ADMIN

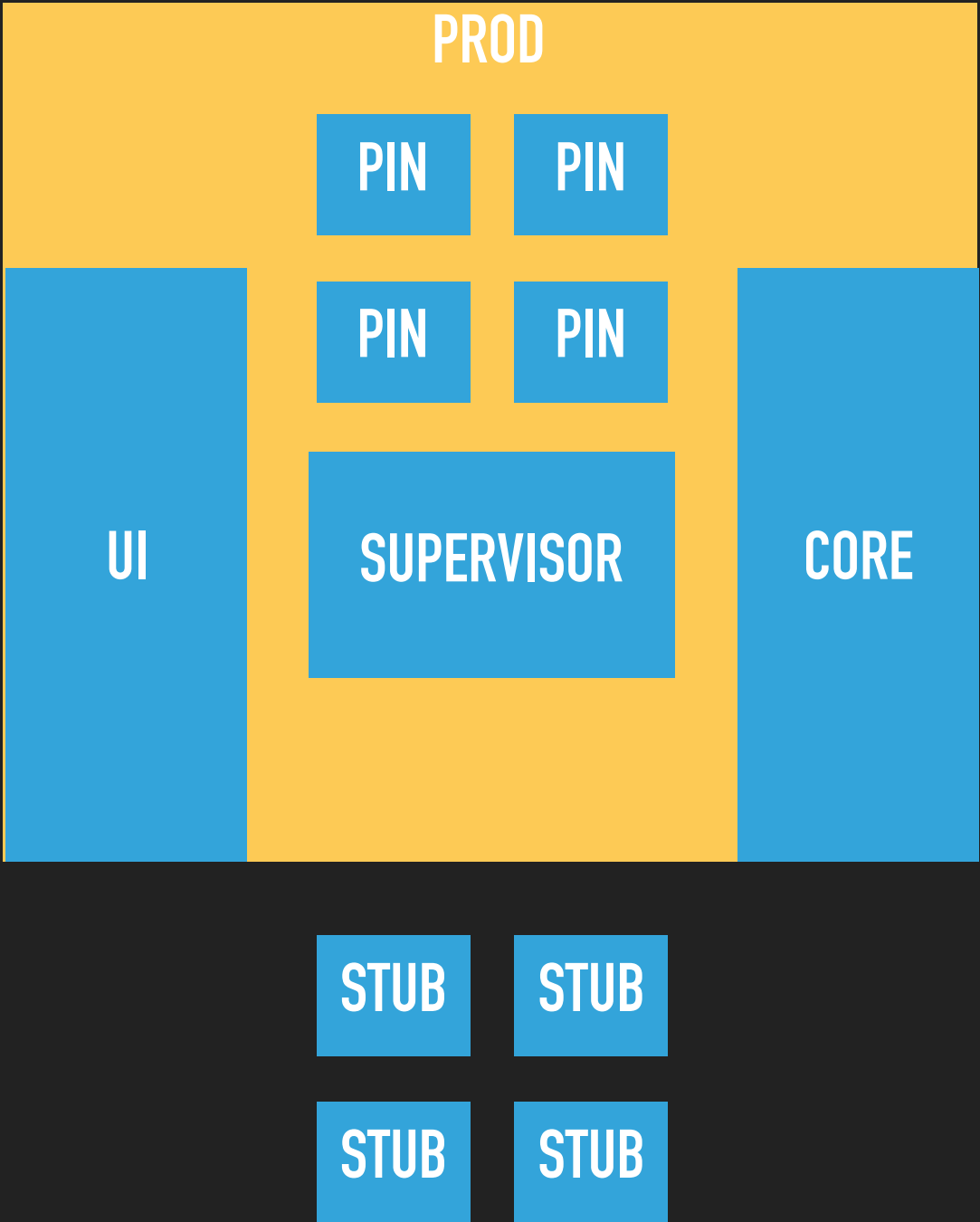
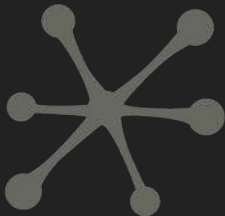
```
your_app_umbrella  
|- your_app_nerves  
|- your_app_ui
```

UI

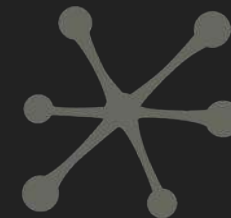
CORE







# GETTING STARTED - USER INTERFACES



Element IO

Home

Configuration

Pinmuxing

Actions

Digital Events

Interval Events

Reboot Device

Download Log

Clear Log

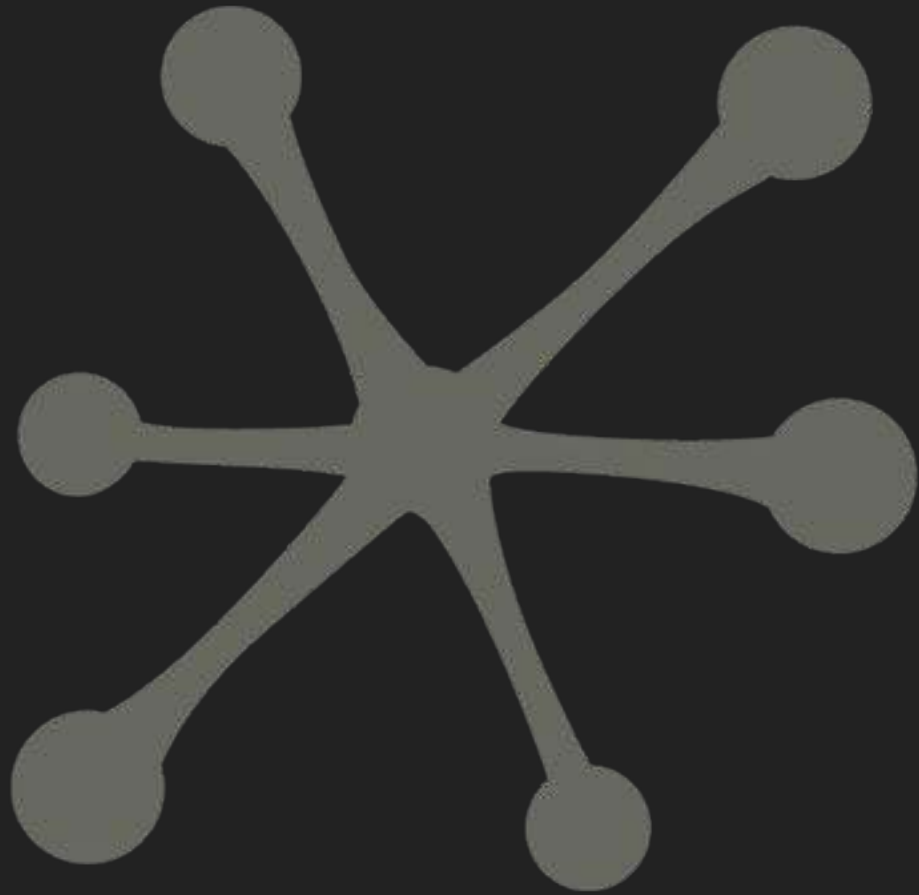


Uptime: 0:0:01:56

Pin 1 ON  
Pin 1 OFF

Counters

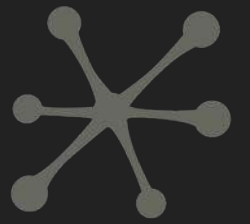
pin 1 monitor: 1



ADVANCED

---

**NERVES FIRMWARE**



## ADDING FILES TO THE ROOT FILE SYSTEM

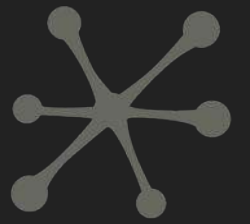
```
config :nerves, :firmware,  
  rootfs_additions: "config/rpi2/rootfs-additions"
```

```
rootfs-additions  
|- etc  
  |- my_utility.conf
```



ROOTFS

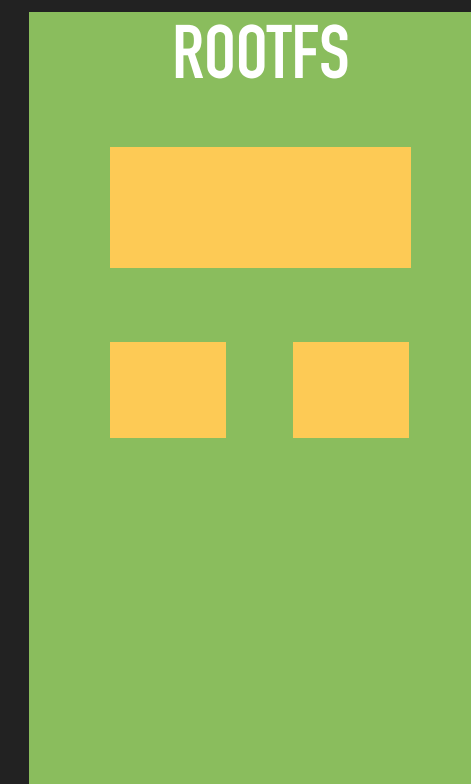


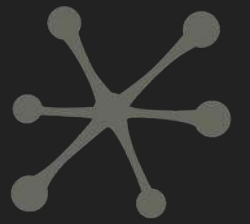


## ADDING FILES TO THE ROOT FILE SYSTEM

```
config :nerves, :firmware,  
  rootfs_additions: "config/rpi2/rootfs-additions"
```

```
rootfs-additions  
|- etc  
  |- my_utility.conf
```





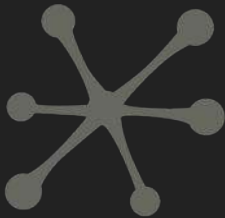
## CHANGING FILES ON ROOT FILESYSTEM

```
config :nerves, :firmware,  
  rootfs_additions: "config/rpi2/rootfs-additions"
```

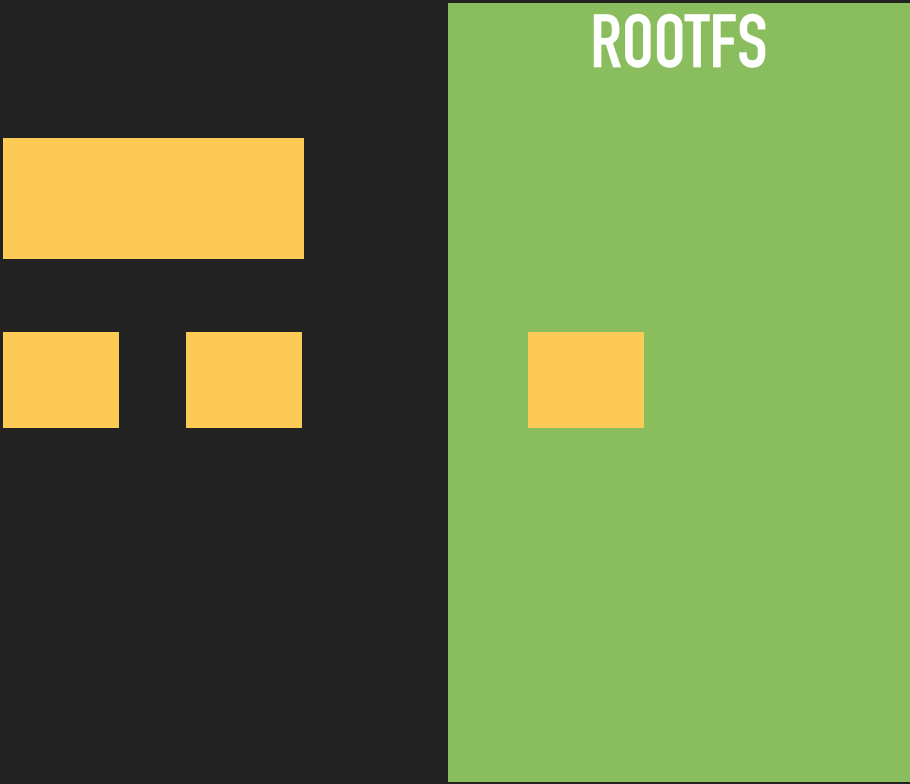
```
rootfs-additions  
|- etc  
  |- erlinit.conf
```

```
# Uncomment to hang the board rather than rebooting when  
Erlang exits  
#--hang-on-exit
```

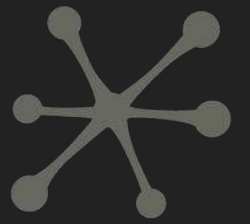




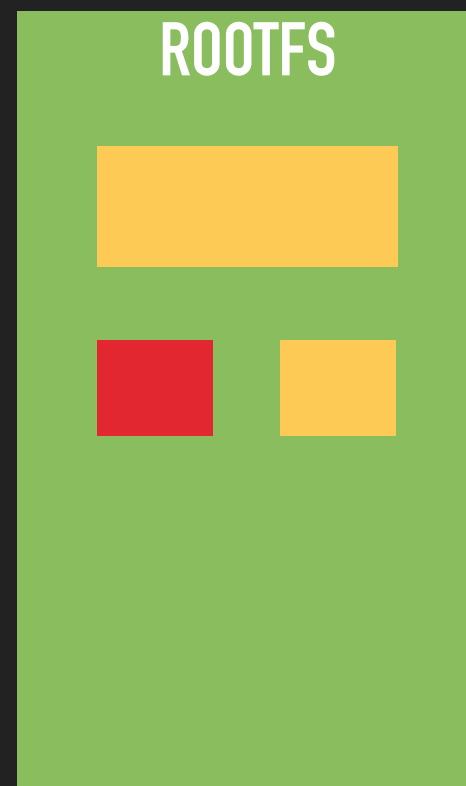
# CHANGING FILES ON ROOT FILESYSTEM

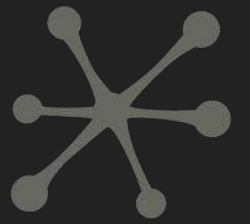






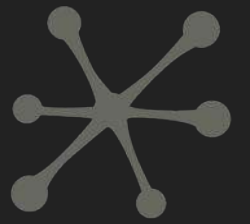
# CHANGING FILES ON ROOT FILESYSTEM





## CHANGING FILES ON BOOT PARTITION



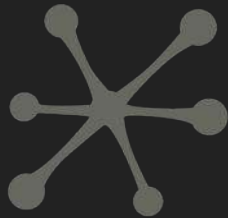


## CHANGING FILES ON BOOT PARTITION

```
# config/rpi2/config.exs
config :nerves, :firmware,
  fwup_conf: "config/rpi2/fwup.conf",
```

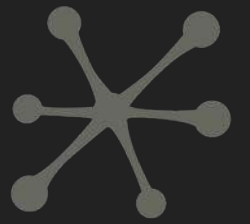
```
# config/rpi2/fwup.conf
file-resource cmdline.txt {
  host-path = "${NERVES_APP}/config/rpi2/cmdline.txt"
}
```

```
# config/rpi2/cmdline.txt
console=tty1 console=ttyS0,115200 root=/dev/mmcblk0p2 rootwait
```



# CHANGING FIRMWARE PARTITIONS

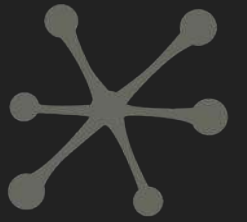
|                            |  |
|----------------------------|--|
|                            |  |
| MBR                        |  |
|                            |  |
| p0: Boot partition (FAT32) |  |
| zImage, bootcode.bin,      |  |
| config.txt, etc.           |  |
|                            |  |
| p1*: Rootfs A (squashfs)   |  |
|                            |  |
| p1*: Rootfs B (squashfs)   |  |
|                            |  |
| p2: Application (FAT32)    |  |
|                            |  |



## CHANGING FIRMWARE PARTITIONS

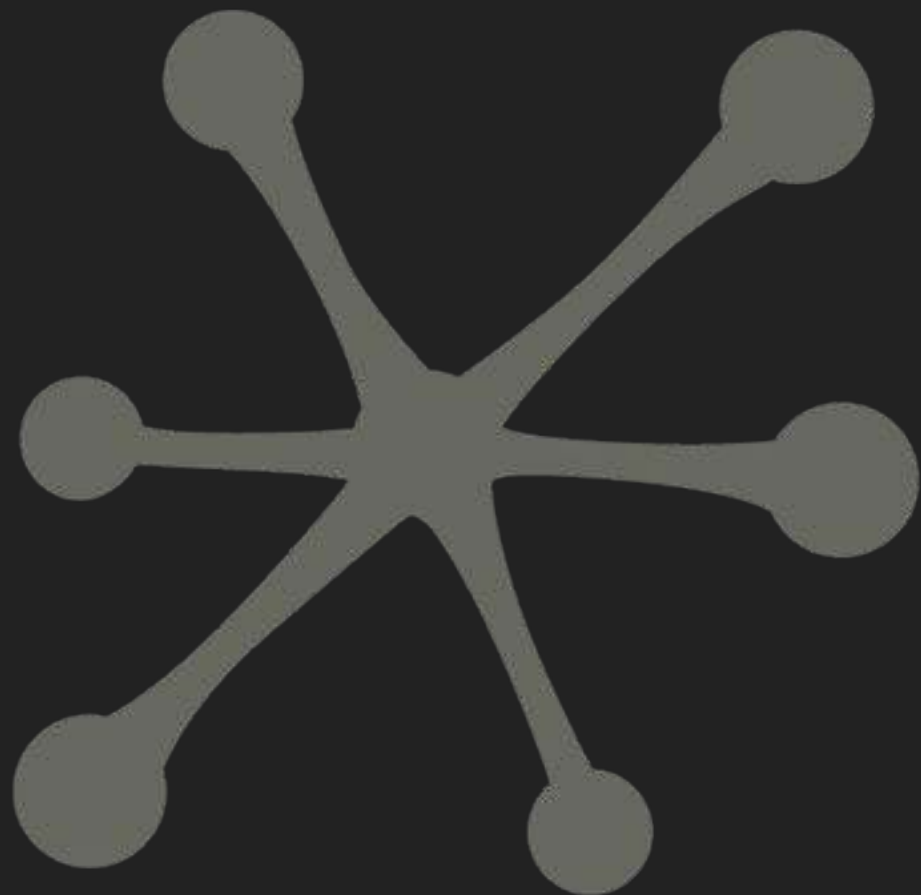
```
# Log partition
define(LOG_PART_OFFSET, 1643048)
define(LOG_PART_COUNT, 1048576)

partition 3 {
    block-offset = ${LOG_PART_OFFSET}
    block-count = ${LOG_PART_COUNT}
    type = 0x83 # Linux
}
```



---

## CONNECTING TO REMOTE NODES

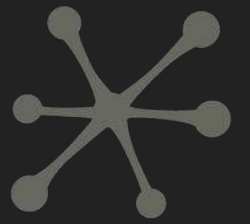


WHATS AHEAD

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**NERVES PROJECT**





## THE FUTURE-ISH

Display / Touch Screen support

Network Firmware Update

Develop Lifecycle

- Always connected target nodes
- Target Distributed ExText
- Development Kits

Easier IO for connecting to arduinos

Video camera support

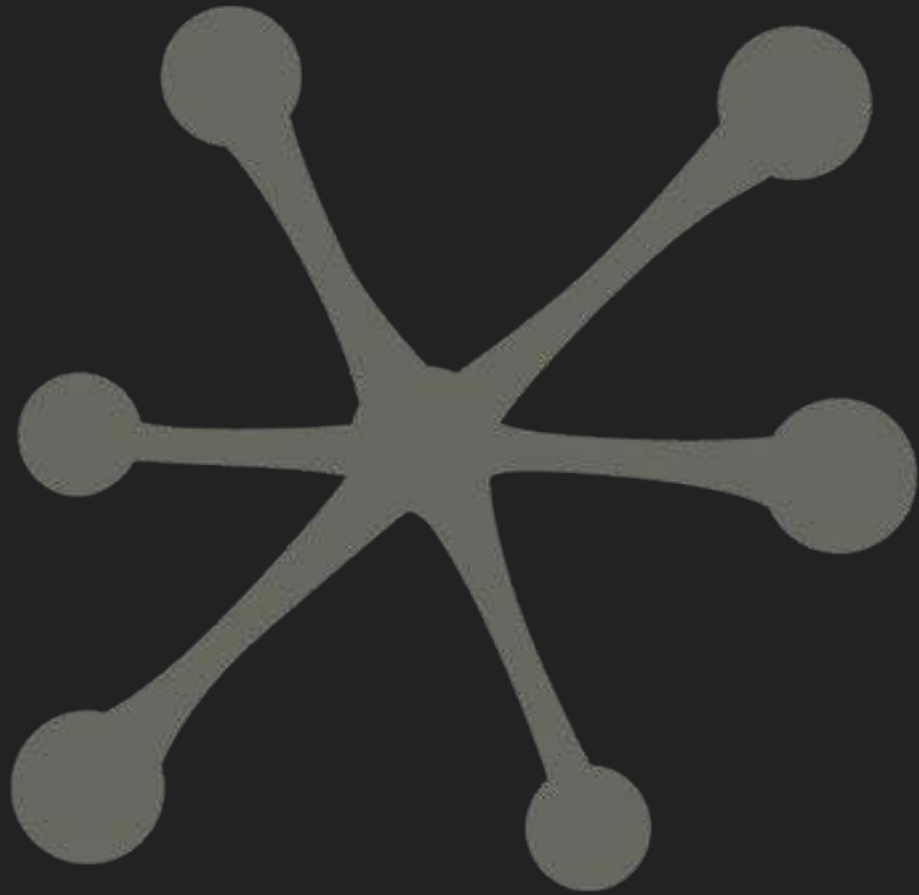
Better Docs

Tutorials

Videos

Books



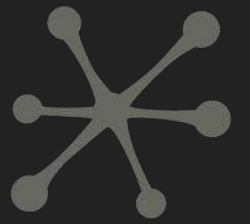


NERVES EMBEDDED SYSTEMS

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**THE REVOLUTION**

THE REVOLUTION



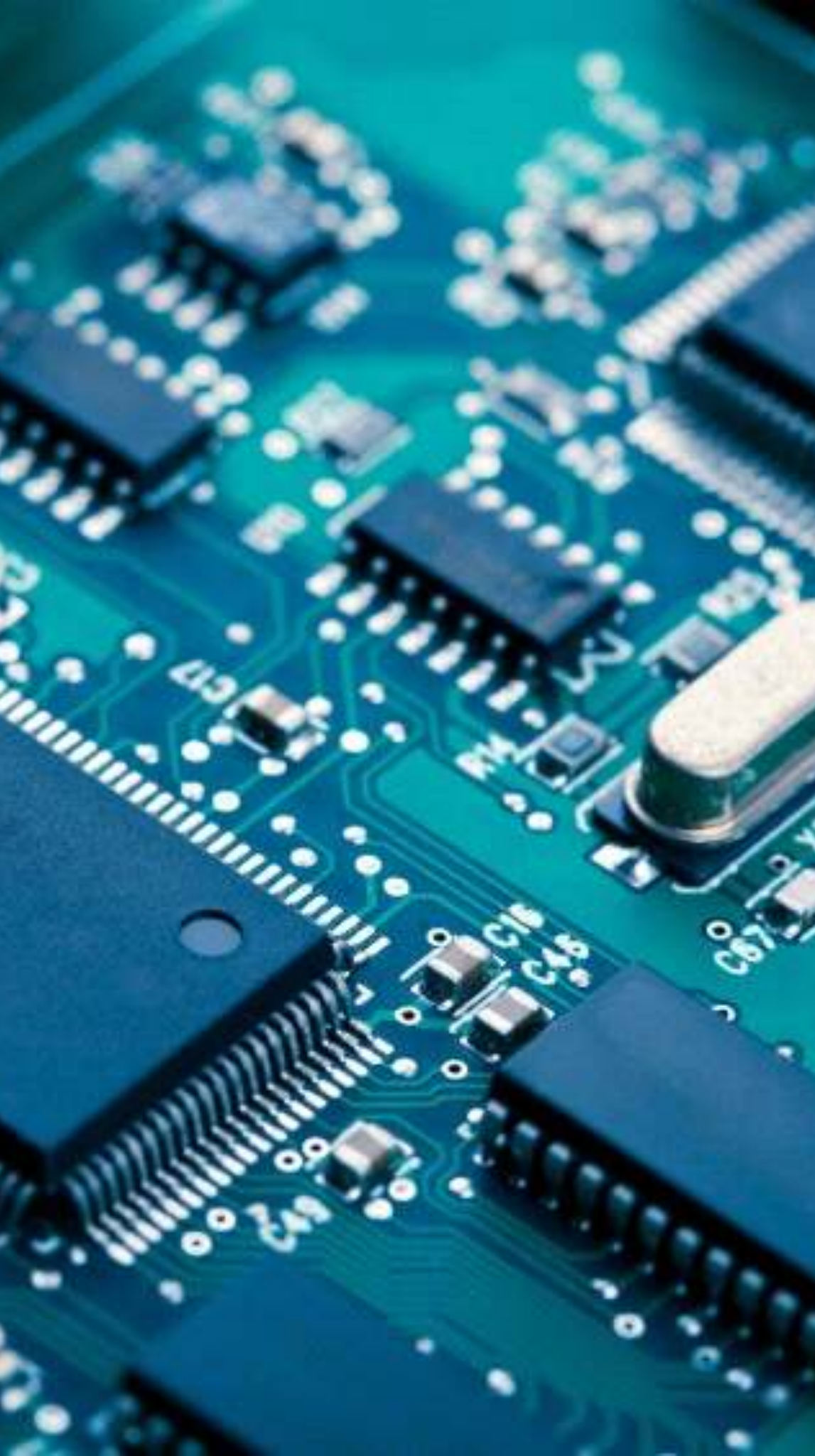
CHANGE THE WORLD



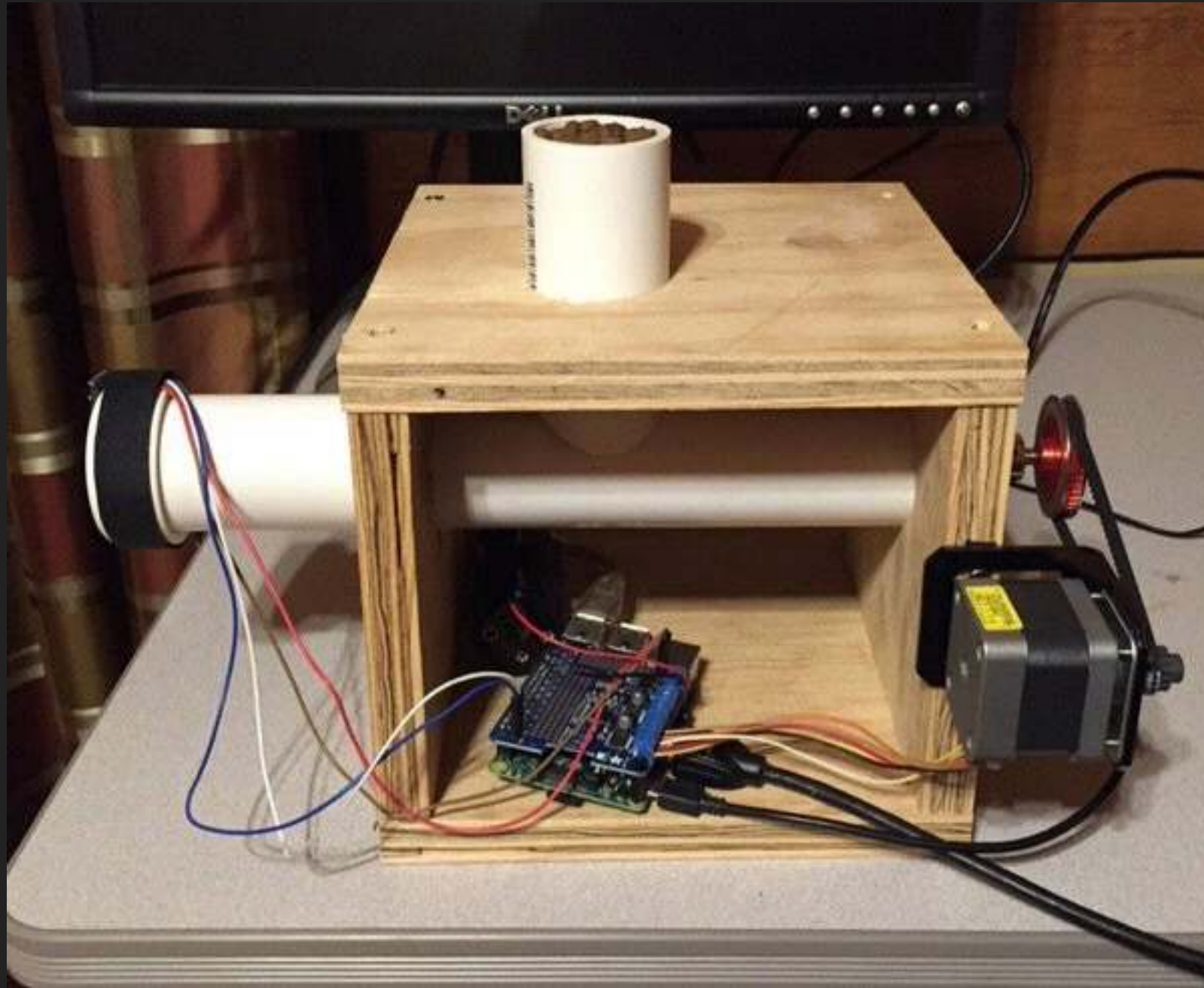
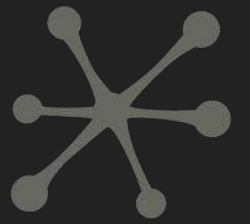








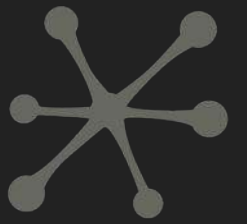
**LETS REWRITE  
EMBEDDED**



@wsmoak

THE REVOLUTION

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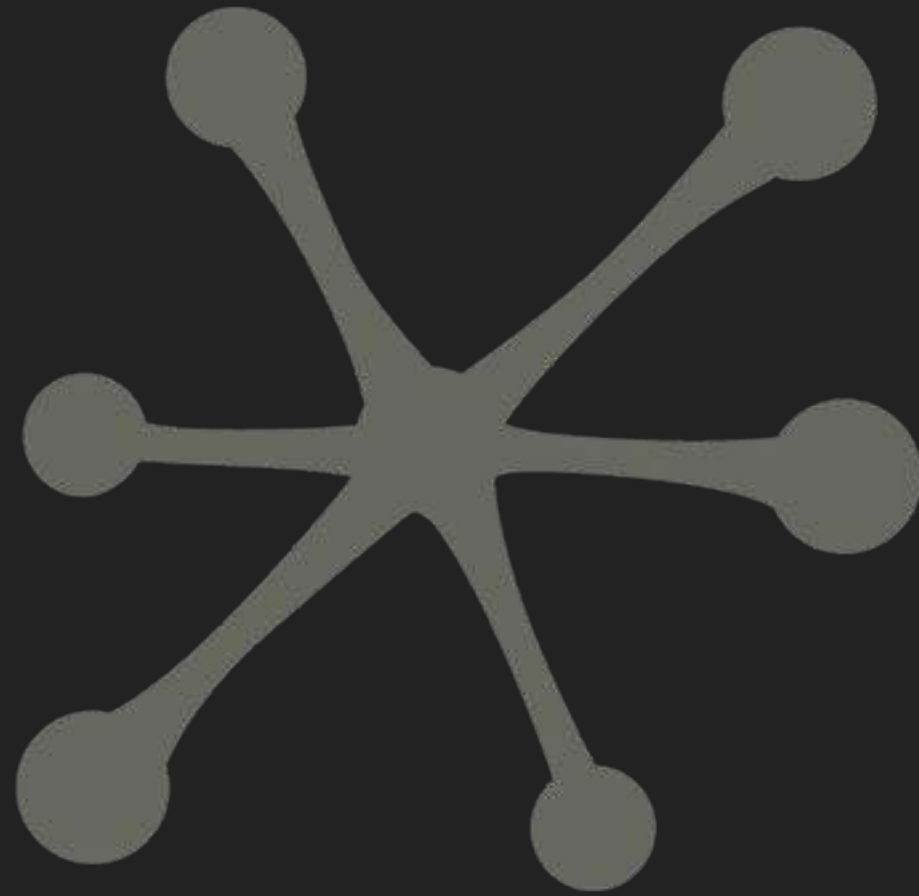


@diptimmo

2016

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Justin Schneck

@mobileoverlord

@nervesproject